Hypothesis Testing for Multiple Population

SDS 320E

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Review: Hypothesis Tests for One Population

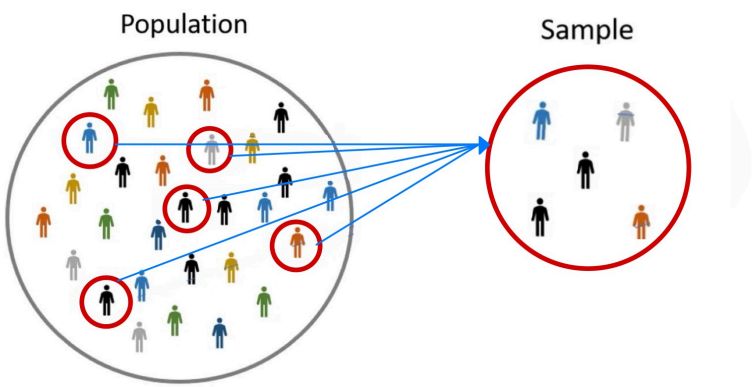
➤Previously, we learned hypothesis tests for one population parameter

Example

Suppose we are interested in determining if the average starting salary of UT Austin students is different from \$72k. Thus, our null and alternative hypothesis are

$$H_0 : \mu = 72,000$$
$$H_a : \mu \neq 72,000.$$

All UT Austin Students Ever Some UT Austin Students



From our sample we get the average starting salary was \$75,000. Thus, we want to know if this is statistically significantly different from \$72,000.

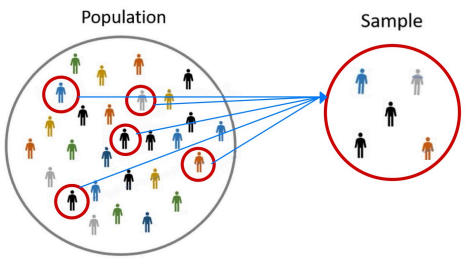
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Hypothesis Testing for Multiple Populations

Example

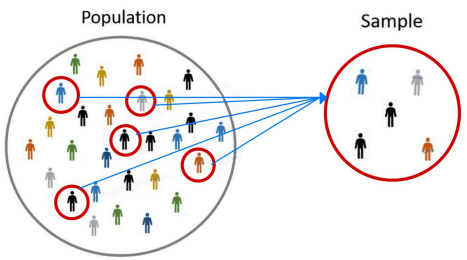
Suppose we are interested in determining if there is a difference among the average starting salary of students that graduated from UT Austin, Texas A&M, and Oklahoma University.

All UT Austin Students Ever — Some UT Austin Students



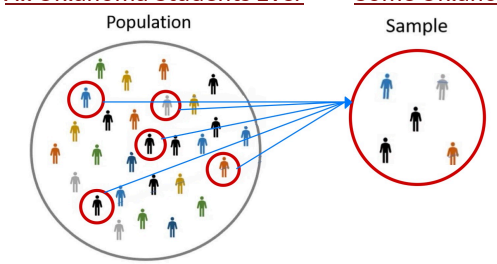
$\longrightarrow \quad \bar{x}_{UT} = \$75k$

All Texas A&M Students Ever — Some Texas A&M Students



$\longrightarrow \quad \bar{x}_{A\&M} = \$73k$

All Oklahoma Students Ever — Some Oklahoma Students



$\longrightarrow \quad \bar{x}_{OU} = \$65k$

Is there a significant difference in salary?

Hypothesis Testing for Two Population Means: Paired & Independent T

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Hypothesis Testing for Two Pop Means

- We may be interested in determining if there is a difference between two means
  - i.e. is $\mu_1 = \mu_2$? μ_1 & μ_2 is mean for population 1 and 2, respectively
- To perform a hypothesis test for μ_1 and μ_2, we must determine:
  - Are the two populations *independent* or *dependent* of one another?
- **Independent Populations:** *populations are independent of one another if they have no influence on, and are completely unrelated to, other populations* i.e. the outcomes from populations are not dependent on one another
- **Dependent Populations:** *populations are dependent on one another if the outcomes from one population influence, or are related to, the outcomes from other populations*
  - Why does dependence matter?

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Why Dependence Matters

Example

Suppose one is interested in determining if there is a difference between μ = *the mean SAT scores* for the genders Male and Female. For H_0, we assume that there is no difference in SAT scores between genders. That is,

$$H_0 : \mu_{Female} = \mu_{Male}$$

Previously, we said p-value $= 0.02 < \alpha = 0.05$. Thus, there is sufficient evidence that there is a significant difference between μ_{Female} and μ_{Male}. However, are there other factors aside from gender that can cause a difference in SAT scores?

Dependent Populations

- There are multiple ways we get dependent populations
  - pre and post treatment from the same subject
  - "pair" subjects based upon similarities most common
  - genetics: twins, litters, ect.

- For dependent populations, to test whether there is a difference, we consider the

X_1 = Dep Pop 1	X_2 = Dep Pop 2
74	70
80	73
68	69
71	76
⋮	⋮

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Hypothesis Test for Two Dependent Means

Paired t-Test

Let X_1 and X_2 represent populations 1 and 2, respectively, and, let $X_d = X_1 - X_2$ and $\mu_d = \mu_1 - \mu_2$. Then, o conduct a hypothesis test for two dependent population means, we will use the *Paired t-Test*:

Null and Alternative Hypothesis

$H_0 : \mu_d = value$ typically $value = 0$, i.e. $\mu_1 = \mu_2$

$H_a : \mu_d < value, > value,$ or $\neq value$

Test-Statistic and Distribution

Assumption for Paired t-Test:

- To do a paired t-Test in R, we will just the `t.test()` function on the differenced data

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Paired t-Test in R

- To conduct a paired t-test in R, we can create the differenced data $X_d = X_1 - X_2$ and then use the `t.test()` function on it. The arguments of interest are:
  - `x`: the numeric vector of the differenced data
  - `mu`: the hypothesized value of μ_d in H_0 (default is 0)
  - `alternative`: the form of H_a, either `"less"`, `"greater"`, or `"two.sided"` (default is `"two.sided"`)
- Or, we can keep our variables separate and use `t.test(x, y, paired = TRUE)`

```r
# Hours of extra sleep for the same 10 patients under drug 1 and drug 2
drug1 <- sleep$extra[sleep$group == 1]
drug2 <- sleep$extra[sleep$group == 2]

# Differenced data: X_d = X_1 - X_2, with X_1 = drug 1 and X_2 = drug 2
diff_data <- drug1 - drug2

# Test H0: mu_d = 0 vs Ha: mu_d > 0 i.e. drug 1 > drug 2
t.test(diff_data, alternative = "greater")
```

```
    One Sample t-test

data:  diff_data
t = -4.0621, df = 9, p-value = 0.9986
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 -2.293005       Inf
sample estimates:
mean of x
    -1.58
```

```r
# Equivalent: let t.test() do the differencing with paired =
t.test(drug1, drug2, paired = TRUE, alternative = "greater")
```

```
    Paired t-test

data:  drug1 and drug2
t = -4.0621, df = 9, p-value = 0.9986
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 -2.293005       Inf
sample estimates:
mean difference
          -1.58
```


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Paired t-Test Example

Example

Suppose one is interested in determining if there is a difference between the mean SAT scores for the genders Female and Male at the 10% significance level. 100 female students were paired to 100 male students based upon their GPA and SAT tutoring and studying hours. The mean difference (Female - Male) for the sample was 9.64, and the standard deviation of the difference was 65.63.

```
1 sat_scores <- read.csv("data/sat_scores.csv")
```

female	male
1036	1019
1069	1059
1021	1008
1075	1130
1034	1066
1155	1009
1103	1074
1113	1087
1086	1079
1151	1035

1. What are the null and alternative hypotheses?

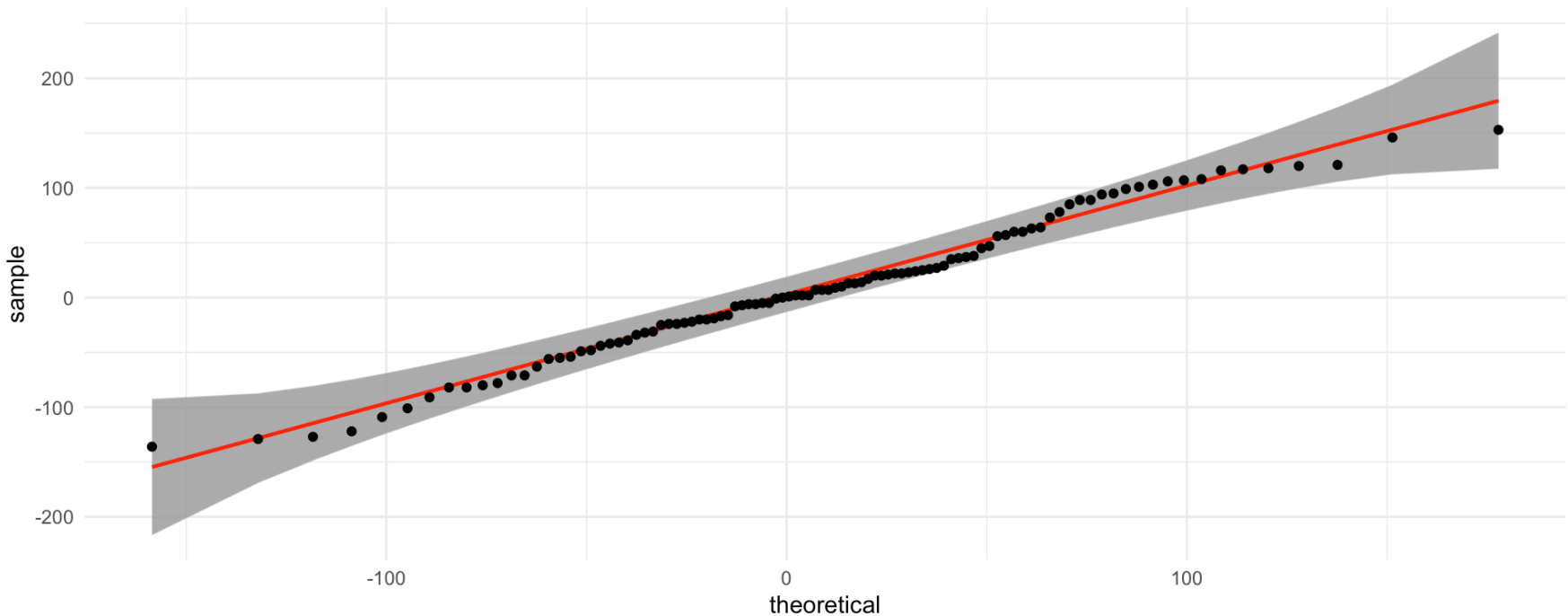
2. What is the Test-Statistic?

Paired t-Test Example: Assumptions

- Before we can perform the paired t-Test, we must check the assumptions for the test

```
1 diff_data <- sat_scores$female - sat_scores$male
```

▶ Show the code



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Paired t-Test Example

Example

Suppose one is interested in determining if there is a difference between the mean SAT scores for the genders Female and Male at the 10% significance level. 100 female students were paired to 100 male students based upon their GPA and SAT tutoring and studying hours. The mean difference (Female - Male) for the sample was 9.64, and the standard deviation of the difference was 65.63.

3. Make a conclusion for the test.

```
1  t.test(diff_data, conf.level = 0.90)
```

```
    One Sample t-test

data:  diff_data
t = 1.4688, df = 99, p-value = 0.1451
alternative hypothesis: true mean is not equal to 0
90 percent confidence interval:
 -1.257667 20.537667
sample estimates:
mean of x
     9.64
```

4. What do we notice about the confidence interval?

Hypothesis Test for Two Independent Means

Pooled t-Test

Let X_1 and X_2 represent populations 1 and 2, respectively, with means μ_1 and μ_2. Then, to conduct a hypothesis test for two independent population means, we will use the *Pooled t-Test*:

Null and Alternative Hypothesis

$H_0 : \mu_1 - \mu_2 = value$ typically $value = 0$, i.e. $\mu_1 = \mu_2$

$H_a : \mu_1 - \mu_2 < value, > value,$ or $\neq value$

Test-Statistic and Distribution

Assumptions for Pooled t-Test

1.
2.

➤ Note: If the equal variance assumption is not reasonable, then we could perform Welch's Unpooled t-Test (less powerful than pooled t-Test)

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Pooled t-Test in R

- To conduct a pooled t-test in R, we can keep the two samples separate and use the `t.test()` function with `var.equal = TRUE`. The arguments of interest are:
  - `x`, `y`: the numeric vectors of the data for population 1 and population 2, respectively
  - `mu`: the hypothesized value of $\mu_1 - \mu_2$ in H_0 (default is 0)
  - `alternative`: the form of H_a, either `"less"`, `"greater"`, or `"two.sided"` (default is `"two.sided"`)
  - `var.equal`: set to `TRUE` for the pooled test (default is `FALSE`, which gives the unpooled test)
- Or, if the data is in one column with a second column of group labels, we can use the formula version `t.test(y ~ group, data = data, var.equal = TRUE)`

```r
# Miles per gallon for cars with automatic and manual transmissions
auto   <- mtcars$mpg[mtcars$am == 0]
manual <- mtcars$mpg[mtcars$am == 1]

# Test H0: mu_1 - mu_2 = 0 vs Ha: mu_1 - mu_2 != 0
# with X_1 = automatic and X_2 = manual
t.test(auto, manual, var.equal = TRUE)
```

```
    Two Sample t-test

data:  auto and manual
t = -4.1061, df = 30, p-value = 0.000285
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -10.84837  -3.64151
sample estimates:
mean of x mean of y
 17.14737  24.39231
```

```r
# Equivalent: formula version, am = 0 (automatic) is group 1
t.test(mpg ~ am, data = mtcars, var.equal = TRUE)
```

```
    Two Sample t-test

data:  mpg by am
t = -4.1061, df = 30, p-value = 0.000285
alternative hypothesis: true difference in means between group 0
and group 1 is not equal to 0
95 percent confidence interval:
 -10.84837  -3.64151
sample estimates:
mean in group 0 mean in group 1
       17.14737        24.39231
```


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Pooled t-Test Example

Example

A researcher is testing a new blood pressure medication. They randomly assigns 29 patients with high blood pressure to one of two groups: 15 receive the new medication and 14 receive a placebo. After 8 weeks, they recorded each patient's reduction in systolic blood pressure (mmHg). They want to know if the new medication reduces blood pressure more than the placebo at the 5% significance level.

```
1 bp_data
```

patient_id	group	reduction_mmHg
6	Treatment	16
11	Treatment	5
10	Treatment	15
1	Treatment	14
13	Treatment	18
27	Placebo	0
26	Placebo	10
28	Placebo	6
17	Placebo	12
19	Placebo	9

1. Are the two populations independent or dependent?

2. What test should we perform?

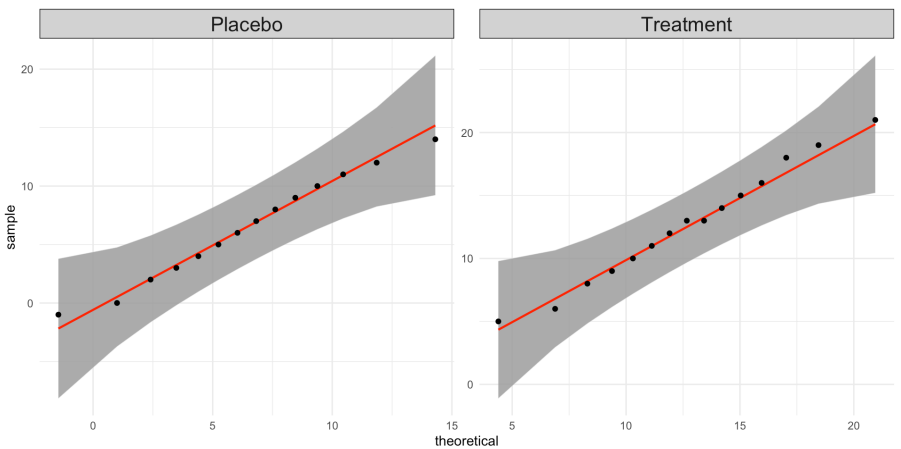
3. What are the null and alternative hypotheses?

Pooled t-Test Example: Assumptions

➤ Before we can proceed with our test, we must check the assumptions first!

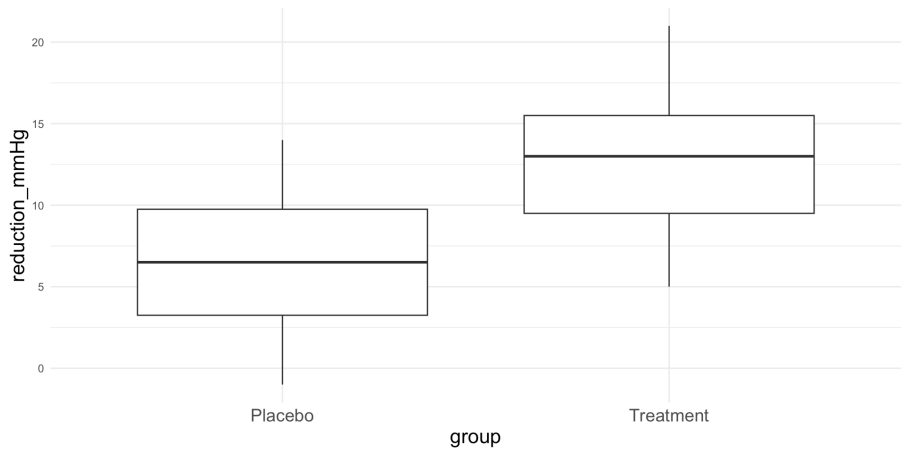
1. Normality

```
library(qqplotr)
ggplot(aes(sample = reduction_mmHg), data = bp_data)+
  stat_qq_band(color = "lightgrey")+
  stat_qq_line(color = "red") +
  stat_qq_point()+
  facet_wrap(~group)
```



2. Equal Variance

```
ggplot(aes(group, reduction_mmHg), data = bp_data)+
  geom_boxplot()
```



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Pooled t-Test Example: Conclusions

Example

A researcher is testing a new blood pressure medication. They randomly assigns 29 patients with high blood pressure to one of two groups: 15 receive the new medication and 14 receive a placebo. After 8 weeks, they recorded each patient's reduction in systolic blood pressure (mmHg). They want to know if the new medication reduces blood pressure more than the placebo at the 5% significance level.

3. Make a conclusion for the test.

```r
treatment <- bp_data$reduction_mmHg[bp_data$group == "Treatment"]
placebo <- bp_data$reduction_mmHg[bp_data$group == "Placebo"]

t.test(treatment, placebo, alternative = "greater", var.equal = TRUE)
```

```
    Two Sample t-test

data:  treatment and placebo
t = 3.6447, df = 27, p-value = 0.000562
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 3.322799      Inf
sample estimates:
mean of x mean of y
12.666667  6.428571
```

4. What is the effect size?

Hypothesis Testing for more than Two Population Means: ANOVA

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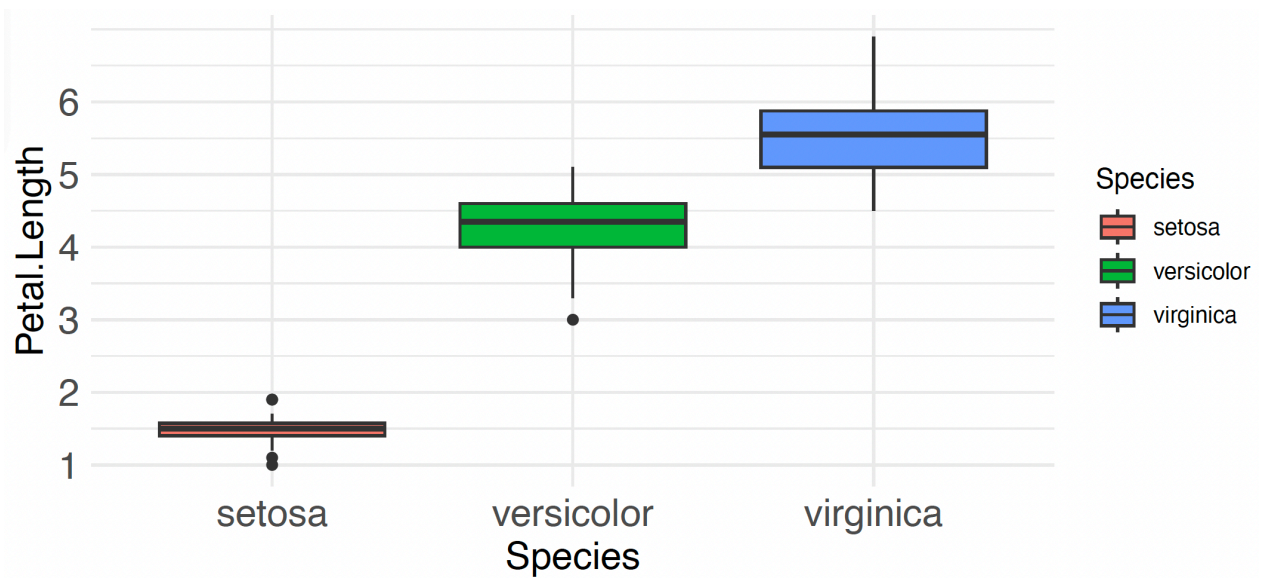
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Test for Multiple Populations

Example

Consider the Iris dataset which consist of petal length of three different flower species: Setos, Versicolor, and Virginica of the Iris flower.

Boxplots | Data



- Visually, we see there is a difference in the typical value among each Species. Now, we will formalize this into a hypothesis test.

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Test for Multiple Populations Cont.

- For testing one and two population means, we often use the t-test
  - For testing more than two population means, we need a different approach
- **Analysis of Variance (ANOVA)** *is a statistical method used to compare the means of three or more groups to determine whether there is a statistically significant difference among them.*
- ANOVA examines how the *dependent/response* variable changes with the levels of the factor(s)
  - A **factor** is a qualitative variable that represents conditions, groups, or categories Ex: Factor is hair color with levels of brown, black, blonde, red, gray
  - The **dependent/response** variable is the quantitative outcome being measured and compared across the different groups.
  - A **one-factor ANOVA** examines how a dependent/response variable changes over the levels of a single factor

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One-Factor ANOVA

One-Factor ANOVA

Let $X_1, \ldots, X_K$ represent *K* different populations and $\mu_k, k = 1, \ldots, K$ are the population means for a specific variable measured across *K* different populations. Then, to conduct a hypothesis test to determine whether the mean is different or the same across the *K* populations, we will use *ANOVA*:

Null and Alternative Hypothesis

$H_0 : \mu_1 = \mu_2 = \ldots = \mu_K$ i.e. the means are all the same

$H_a :$ At least one mean μ is different.

Test-Statistic and Distribution

$$F = \frac{\sum_{k=1}^{K} n_k(\bar{X}_k - \bar{X})^2/(K-1)}{\sum_{k=1}^{K}\sum_{i=1}^{n_k}(X_i - \bar{X}_k)^2/(n-K)} \sim F(df_1 = K-1, df_2 = n-K)$$

F is another distribution like t, normal, ect.

Assumptions for ANOVA

1.

2.

Between vs With Variability

Idea

The main idea of *ANOVA* is to measure the variation between groups relative to the variation within groups.

Table 1. Sepal.Width measurement for three Iris Species

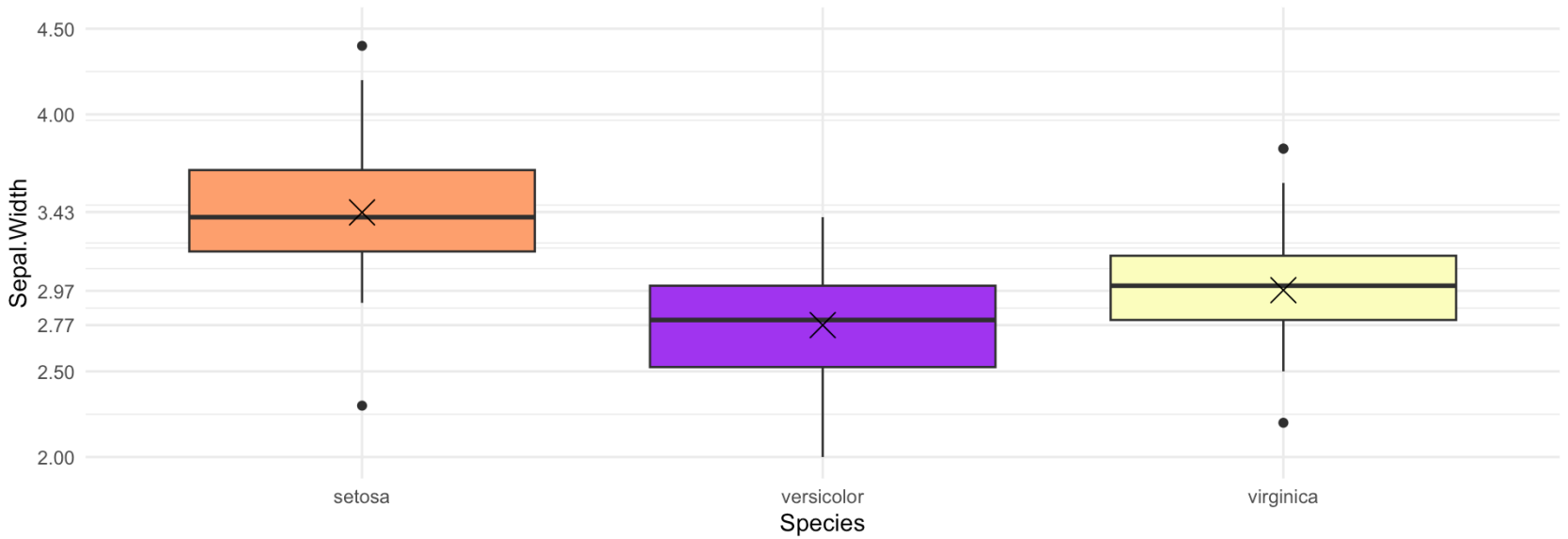
	setosa	versicolor	virginica
	3.5	3.2	3.3
	3.0	3.2	2.7
	3.2	3.1	3.0
	3.1	2.3	2.9
	3.6	2.8	3.0
	3.9	2.8	3.0
Mean	**3.38**	**2.90**	**3.15**

- Between group variability refers to the variation of the group means relative to the overall mean i.e. how far apart the groups are from each other
- Within group variability refers to the variation of the measurement values inside each group i.e. how spread out the observations are within that group

Between vs Within Illustrated

Example

Consider the width of three different flower species: Setos, Versicolor, and Virginica.



- There is some *between* group variability among Versicolor and Virginica means are slightly different
  - However, when we consider the *within* group variability, the difference is not significant
- Once we include Setosa, the *between* group variability is large enough compared to the *within* group variability. Thus, the difference is significant i.e at least one mean is different than the rest

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ANOVA in R

- To perform a one-factor ANOVA test in R we use the `aov()` function

```
aov(response ~ factor, data = my_data)
```

- To summarize the results of our ANOVA test (p-value and ect…) we use the `summary()` function on our `aov` object

```
aov_results <- aov(response ~ factor, data = my_data)
summary(aov_results)
```

Example

Consider the previous example comparing the width among the Species for the iris dataset.

```
aov_results <- aov(Sepal.Width ~ Species, data = iris)
summary(aov_results)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
Species       2  11.35   5.672   49.16 <2e-16 ***
Residuals   147  16.96   0.115
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


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ANOVA Example

Example

Suppose Starbucks wants to compare the effectiveness of different advertising channels. Starbucks selects 40 locations in the US to get TV adds only, 40 locations to get online adds, and 40 locations to get print ads. They record the sales increase percentage for all locations and want to compare the mean sales increase across the advertising methods. Using one-factor ANOVA, make a conclusion about the average sales increase across advertising methods at the 5% significance level.

```
1 starbucks
```

Data | Means

sales	method
6.0760387	TV
1.6916368	TV
3.8594772	TV
6.4265286	Online
3.2314372	Online
2.8701607	Online
0.8012964	Print
5.3018618	Print
0.4740863	Print

1. What test should we perform?

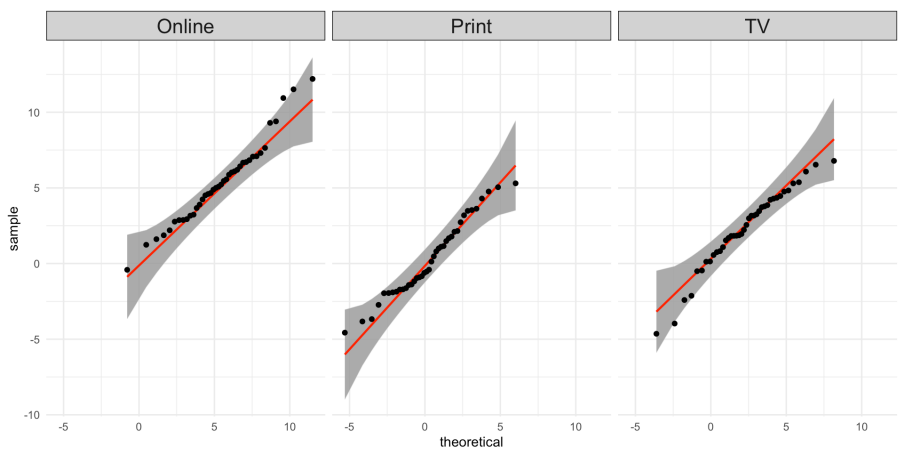
2. What are the null and alternative hypotheses?

ANOVA Example: Assumptions

➤ Before we can proceed with our test, we must check the assumptions first!

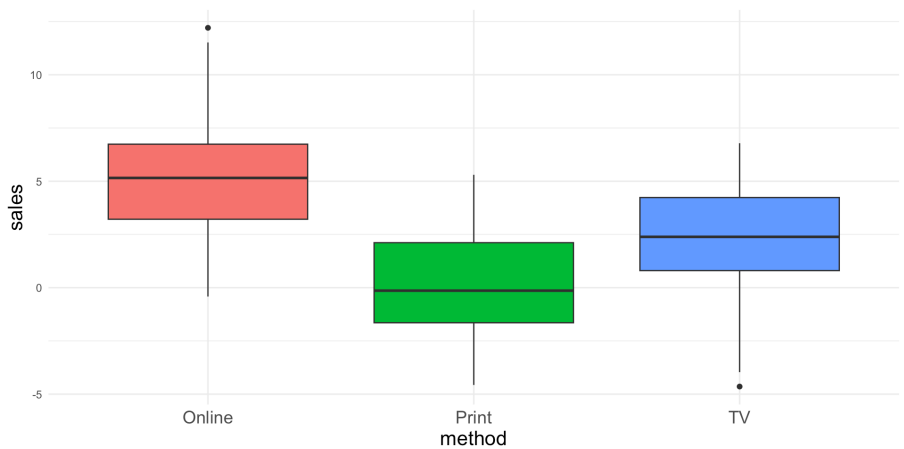
1. Normality

```
library("qqplotr")
starbucks |>
ggplot(aes(sample = sales))+
  stat_qq_band(color = "lightgrey")+
  stat_qq_line(color = "red") +
  stat_qq_point()+
  facet_wrap(~method)
```



2. Equal Variance

```
ggplot(aes(method, sales, fill = method), data = starbucks)+
  geom_boxplot()
```



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ANOVA Example: Conclusion

Example

Suppose Starbucks wants to compare the effectiveness of different advertising channels. Starbucks selects 40 locations in the US to get TV adds only, 40 locations to get online adds, and 40 locations to get print ads. They record the sales increase percentage for all locations and want to compare the mean sales increase across the advertising methods. Using one-factor ANOVA, make a conclusion about the average sales increase across advertising methods at the 5% significance level.

3. Make a conclusion for the test.

```r
aov_results <- aov(sales ~ method, data = starbucks)
summary(aov_results)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
method        2    511  255.51   35.93 6.84e-13 ***
Residuals   117    832    7.11
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Results:
  - Between Variability (MSB) =
  - Within Variability (MSW) =
  - F-Statistic =
  -

Conclusion:

Multiple Comparisons for ANOVA

Reminder

The hypotheses for ANOVA are

$$H_0 : \mu_1 = \mu_2 = \mu_3$$
$$H_a : \text{At least one mean } \mu \text{ is different,}$$

- Thus, if p-value $< \alpha$, we reject H_0 and conclude we have sufficient evidence that
  - So, the question becomes which one(s) is/are different from one another?
- The goal of **multiple comparison** analysis is to determine which mean(s) is different from the rest sometimes all of the means are different

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Multiple Comparisons for ANOVA Cont.

- We could answer this by performing pairwise hypothesis tests i.e. test each possible pair of groups

$$H_0 : \mu_1 = \mu_2 \quad H_0 : \mu_1 = \mu_3 \quad H_0 : \mu_2 = \mu_3$$
$$H_a : \mu_1 \neq \mu_2 \quad H_a : \mu_1 \neq \mu_3 \quad H_a : \mu_2 \neq \mu_3$$

- Intuitively this method is valid, however, statistically it creates a problem

Issue

When we run many hypothesis tests, we increase the chance of finding a "significant" difference just by "luck".

Reminder: In hypothesis testing, we never know if our conclusion is correct

- Now, imagine if we have 100 means to compare…

$$H_0 : \mu_1 = \mu_2 \quad \ldots \quad H_0 : \mu_{97} = \mu_{98} \quad H_0 : \mu_{97} = \mu_{99} \quad H_0 : \mu_{97} = \mu_{100}$$
$$H_a : \mu_1 \neq \mu_2 \quad \ldots \quad H_a : \mu_{97} \neq \mu_{98} \quad H_a : \mu_{97} \neq \mu_{99} \quad H_a : \mu_{97} \neq \mu_{100}$$

- That leaves a lot of chance for error, which we call the **family-wise error rate (FWER)**

$$FWER = P(\text{at least one error}) = 1 - (1 - \alpha)^{\#\text{ of test}}$$

This is a lot bigger than our original α!

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Tukey's Multiple Comparison Adjustment

- We need a way to adjust for this inflation of error before drawing conclusions

Tukey's Multiple Comparison

After rejecting H_0 in ANOVA, we use *Tukey's Multiple Comparison* to determine which pairs of means are different, while keeping the family-wise error rate at α:

Null and Alternative Hypothesis

$H_0 : \mu_i = \mu_j$ for every pair of groups $i \neq j$

$H_a : \mu_i \neq \mu_j$

Conclusions

- If the adjusted p-value $< \alpha$,
- Or, if the interval for $\mu_i - \mu_j$ does not contain 0,

- To perform Tukey's Multiple Comparison in R, we use `TukeyHSD()` with `aov()`

```
aov_results <- aov(response ~ factor, data = my_data)
TukeyHSD(aov_results)
```

Tukey Multiple Comparison Example

Starbucks Example Revisited

We are wanting to compare advertising strategies for Starbucks between online, print, and tv ads at the 5% significance level. We rejected H_0. Thus, we know at least one mean is signficantly different from another.

➤ Now, we determine which mean is different using Tukey's Multiple Comparisons

```
1 aov_results <- aov(sales ~ method, data = starbucks)
2 TukeyHSD(aov_results)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = sales ~ method, data = starbucks)

$method
                  diff        lwr       upr     p adj
Print-Online -5.010282 -6.4258489 -3.594714 0.0000000
TV-Online    -3.084722 -4.5002892 -1.669155 0.0000029
TV-Print      1.925560  0.5099923  3.341127 0.0045610
```

➤ Print vs Online:

➤ TV vs Online:

➤ TV vs Print:

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Tukey's Multiple Comparison Example Cont.

- Using the p-values from Tukey's Multiple comparisons tells us which means are different
  - However, we also want to know how different the means are. Thus, we look at the Tukey's Confidence Intervals

```
1 TukeyHSD(aov_results)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = sales ~ method, data = starbucks)

$method
                  diff        lwr       upr     p adj
Print-Online -5.010282 -6.4258489 -3.594714 0.0000000
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TV-Print      1.925560  0.5099923  3.341127 0.0045610
```

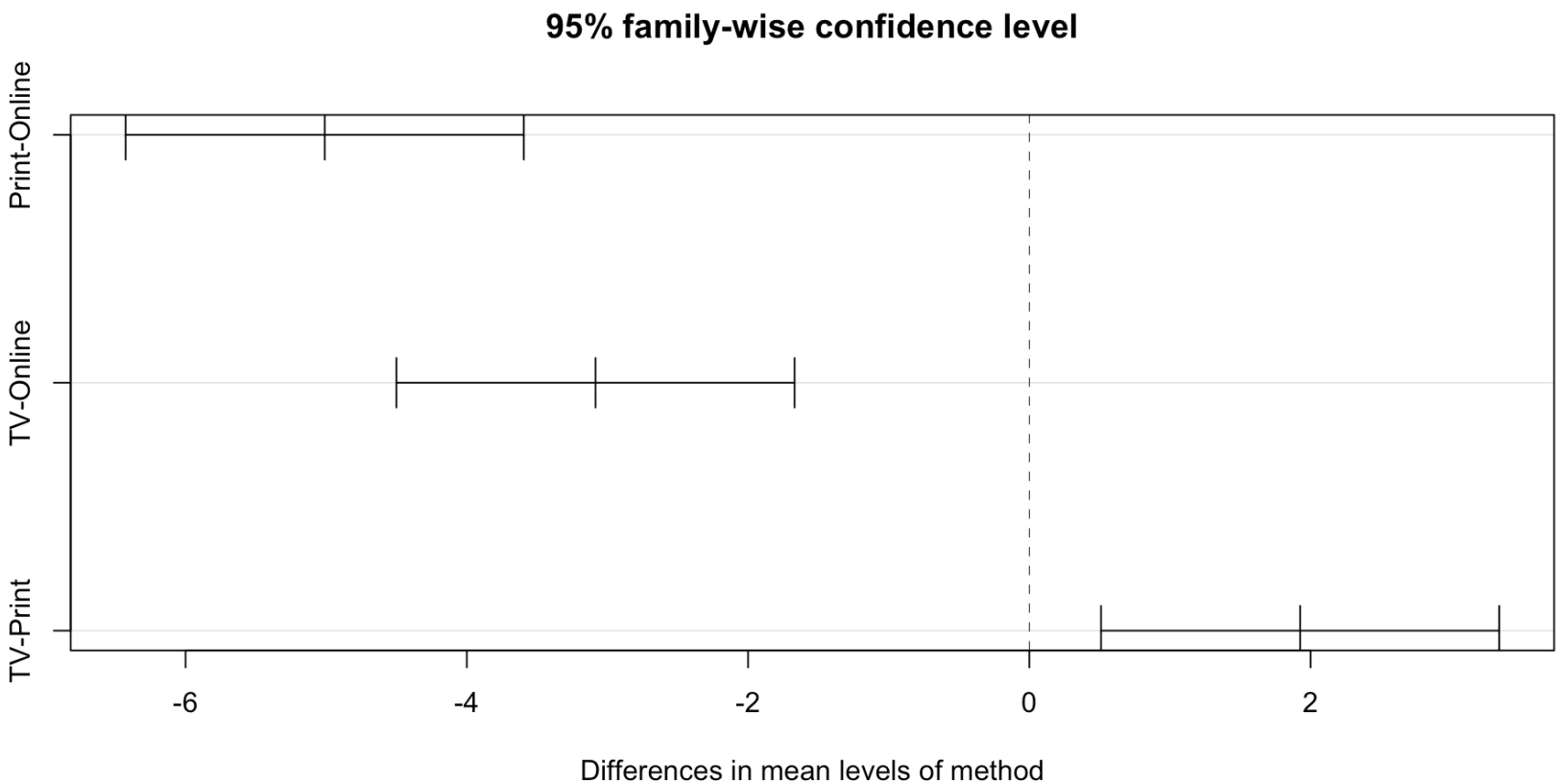
- Print vs Online: Print sales increase is 6.42% to 3.59% lower than Online, on average
- TV vs Online: TV sales increase is 4.50% to 1.67% lower than Online, on average
- TV - Print: TV sales increase is 0.51% to 3.34% higher than Print, on average

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Tukey's Multiple Comparison Example Cont.

- We can also plot the confidence intervals

```
1 Tukey_results <- TukeyHSD(aov_results)
2 plot(Tukey_results)
```



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Hypothesis Testing for Categorical Data: χ^2 GOF

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Chi-Square Goodness of Fit Test

Chi-Square Goodness of Fit Test

Let a categorical variable have *K* categories, and let $p_k, k = 1, \ldots, K$ be the population proportions of each category. Then, to conduct a hypothesis test to determine whether the proportions match a hypothesized distribution, we will use the *Chi-Square Goodness of Fit Test*:

Null and Alternative Hypothesis

$H_0 : p_1 = p_{1,0},\ p_2 = p_{2,0},\ \ldots,\ p_K = p_{K,0}$ sometimes $p_1 = p_2 = \ldots = p_K = 1/K$

H_a : At least one proportion p_k is different from its hypothesized value.

Test-Statistic and Distribution

O_k = observed count, $E_k = n \cdot p_{k,0}$ = expected count if H_0 is true

Assumptions for Chi-Square Goodness of Fit

1.

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Chi-Square Goodness of Fit Test in R

- To conduct a chi-square goodness of fit test in R, we can use the `chisq.test()` function on the observed counts. The arguments of interest are:
  - `x`: the numeric vector of the observed counts O_k for each category
  - `p`: the vector of hypothesized proportions $p_{k,0}$ in H_0, in the same order as `x` (default is equal proportions)
- Or, if we have the raw data, we can first use `table()` to get the counts and then use `chisq.test(table(data), p = p)`
- To check the assumption that all expected counts are at least 5, we can use `$expected`

```r
# Number of cars with 4, 6, and 8 cylinders
observed <- c(11, 7, 14)

# Test H0: p_4 = 0.4, p_6 = 0.2, p_8 = 0.4
# vs Ha: at least one proportion is different
chisq.test(x = observed, p = c(0.4, 0.2, 0.4))
```

```
    Chi-squared test for given probabilities

data:  observed
X-squared = 0.42187, df = 2, p-value = 0.8098
```

```r
# Equivalent: start from the raw data and use table() for the
chisq.test(table(mtcars$cyl), p = c(0.4, 0.2, 0.4))
```

```
    Chi-squared test for given probabilities

data:  table(mtcars$cyl)
X-squared = 0.42187, df = 2, p-value = 0.8098
```


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Election Example

Example

The latest Emerson College poll for the Texas Senate race polled 947 decided voters that were likely to vote. Of the 947, 476 said they would vote for James Talarico (D), 458 for Ken Paxton (R), and 13 for Ted Brown (L). We want to make a conclusion at the 5% level of significance on whether there is a significant difference in the proportion of likely voters who support each of the three candidates.

1. What test should we perform?

2. What are the null and alternative hypothesis?

3. Are the assumptions for the Test met?

Election Example Continued

Example

The latest Emerson College poll for the Texas Senate race polled 947 decided voters that were likely to vote. Of the 947, 476 said they would vote for James Talarico (D), 458 for Ken Paxton (R), and 13 for Ted Brown (L). We want to make a conclusion at the 5% level of significance on whether there is a significant difference in the proportion of likely voters who support each of the three candidates.

4. Make a conclusion for the test.

```r
observed <- c("Talarico" = 476, "Ken Paxton" = 458, "Brown" = 13)

chisq.test(observed, p = c(1/3, 1/3, 1/3))
```

```
    Chi-squared test for given probabilities

data:  observed
X-squared = 435.82, df = 2, p-value < 2.2e-16
```

5. From the poll, Talarico "had" 476 votes > Ken Paxton's 458 votes. Does our conclusion support that Talarico will win?

Bonferroni Confidence Intervals

- Since we rejected H_0, the natural question is: which proportion(s) are different from the hypothesized value?
  - We can answer this by estimating each proportion with a confidence interval
- However, just like multiple hypothesis tests,

Bonferroni Confidence Intervals

To build K confidence intervals while keeping the overall level of significance at α, we use the *Bonferroni adjustment*:

We will split α across the K different confidence intervals (i.e. each interval gets α/K instead of α) and construct K intervals at level of confidence.

Bonferroni CI in R: Copy, paste, and save this function to your environment in R.

```r
bonferroni_ci <- function(observed, alpha = 0.05) {
  K <- length(observed)
  n <- sum(observed)
  lapply(observed, function(x) {
    prop.test(x, n, conf.level = 1 - alpha / K, correct = FALSE)$conf.int
  })
}
```

- `observed`: the numeric vector of the observed counts for each category
- `alpha`: the overall (family-wise) level of significance (default is 0.05)

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Election Example Cont.

Example

The latest Emerson College poll for the Texas Senate race polled 947 decided voters that were likely to vote. Of the 947, 476 said they would vote for James Talarico (D), 458 for Ken Paxton (R), and 13 for Ted Brown (L). We want to make a conclusion at the 5% level of significance on whether there is a significant difference in the proportion of likely voters who support each of the three candidates.

6. Since we rejected H_0, which candidate(s) have a proportion different from assumed?

```
observed <- c(476, 458, 13)
bonferroni_ci(observed, alpha = 0.05)
```

```
[[1]]
[1] 0.4638448 0.5414033
attr(,"conf.level")
[1] 0.9833333

[[2]]
[1] 0.4449719 0.5224901
attr(,"conf.level")
[1] 0.9833333

[[3]]
[1] 0.007165844 0.026139606
attr(,"conf.level")
[1] 0.9833333
```

- CI for Talarico: (46.38%, 54.14%).
- CI for Paxton: (44.50%, 52.25%).
- CI for Brown: (0.72%, 2.61%)

Note: This tells us which proportions are different than assumed. But, we can not make a conclusion on whether Talarico or Paxton will win.

Bonferroni Confidence Intervals: Multiple Comparisons

- Since we rejected H_0, we know that the candidates do not have equal support.
  - But can we conclude that one candidate is more likely to win?
- To answer this, we compare the population proportions for each pair of candidates.
  - We construct bonferroni confidence intervals for the differences between proportions, $p_i - p_j$.

More math and statistics that is beyond this course…

Comparison	Lower_CI	Upper_CI
Talarico vs. Ken Paxton	-0.0582365	0.0962513
Talarico vs. Brown	0.4479443	0.5298804
Ken Paxton vs. Brown	0.4289947	0.5108152

- The interval for Talarico vs Ken Paxton contains 0. Thus, we can not conclude there is a significant difference between Talarico and Ken Paxton i.e., we can not conclude who will win.

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The End